
Part II

The Interference Effect



Proposed Mechanisms Underlying the Interference Effect

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Introduction to the Interference Effect

In 1980, Dr. Hickson [1] published a seminal paper on concurrent training, comparing the effects of endurance training and strength training alone with those of concurrent training on aerobic capacity and leg strength in humans [1]. Since then, the assumption has been that there is an interference effect between the two training modalities, also known as the concurrent training, or interference, effect [2]. The interference effect can be further broken down into the acute interference effect, whereby the residual fatigue from the first bout decreases the subsequent performance of the second bout [3], and the chronic effect (the focus of this review), whereby the adaptation to endurance or strength is modified by the participation in the other form of training [4]. In short, performing the two exercise modalities in proximity of each other over an extended period of time may decrease the effects of strength training on muscle strength [1], mass [5], and power [4], with no apparent negative effects on outcomes of endurance training [1], which indeed is likely to benefit from concurrent training [6]. Animal studies have consistently attributed this impairment to activation and deactivation of specific molecular switches and cellular signaling pathways [2, 7]. Over the decades, it has become increasingly clear that the universality of the interference effect is questionable in humans, with an

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increasing bulk of evidence suggesting that its presence and extent depends on training status, type of muscle being trained, endurance training modality and protocol [2, 7]. This may explain data from human muscle biopsies, which often end up negative for proxy markers of an interference effect [8, 9].

In a meta-analysis from 2012 [4], Wilson and colleagues attempted to draw a definitive conclusion from the human studies investigating the interference effects of concurrent training. Their conclusions were that concurrent training leads to increases in strength and muscle mass that were not statistically less than strength training alone, suggesting that endurance training does not impose a generic interference effect on strength adaptations [4]. The analysis only reached statistical significance when looking at muscle power, which was lower in the concurrent group than those who completed strength alone [4]. In a subset of analyses, different modes of endurance training were associated with different degrees of interference, with running but not cycling impairing increases in lower body strength and muscle mass. Lastly, the degree of interference was associated with the volume of endurance training, showing a negative correlation with both frequency and average duration of endurance workout. In a sense, Wilson et al. [4] summarized what we already suspected: if you train at a high enough frequency, intensity, and duration, endurance exercise inhibits strength development. Therefore, in high-level athletes, an interference may exist, but in recreational athletes who only train 3–6 times a week, the effect seems to be equivocal [10]. Further, since the response to training varies with age, gender, training status, and genetics, observing a statistically significant concurrent training effect in a small cohort is difficult. By contrast, when training >6 times a week at high intensities, there is a decrease in both strength [1] and hypertrophy [5] when strength and endurance training are performed together.

The Interference Effect and Hypertrophy

The molecular mechanism underlying the interference effect has been an area of research interest for almost 15 years [11]. By 1999, complex 1 of the mechanistic target of rapamycin (mTORC1) had been identified as one of the key molecular components of the hypertrophic response to resistance exercise [12]. In 2003, the AMP-activated protein kinase (AMPK) was shown to directly inactivate mTORC1 and limit growth in cells [11]. Since AMPK had previously been shown to be activated by endurance exercise [13], the logical hypothesis was that when activated by endurance exercise, AMPK would shut down mTORC1 and reduce muscle hypertrophy. In agreement with this hypothesis, old animals that showed increased AMPK activity hypertrophied less following overload [14] and the pharmacological AMPK activator 5-aminoimidazole-4-carboxamide-1- β -ribofuranoside (AICAR) could prevent the activation of mTORC1 following resistance exercise [15]. These data seemed to confirm that the concurrent training effect was the result of a simple molecular switch: strength exercise activated mTORC1, endurance activated AMPK, and when performed together AMPK shut of mTORC1 reduced muscle hypertrophy [16].

Unfortunately, this simple relationship did not translate into human muscle. When Apro and colleagues began to look for an acute inhibition of mTORC1

signaling in human muscle after concurrent training they found that neither steady-state cycling for 30 min at 70% of VO_2max [9], nor five 4-min intervals at 85% of VO_2max with 3 min rest in between [8] inhibited the activation of mTORC1 following resistance exercise in moderate trained persons. The high intensity interval training, which activated AMPK, did decrease mTORC1 activity marginally, but this did not reach statistical significance, suggesting that a simple relationship where AMPK activation during endurance exercise blocked mTORC1 activation by resistance exercise was not the mechanism underlying the interference effect.

In hindsight, there were many clues that this simple mechanism was not sufficient. First, in models of concurrent training, such as overload hypertrophy in rodents, it is the $\alpha 1$ isoform of AMPK that is upregulated [17] and only when $\alpha 1$ AMPK is knocked out does skeletal muscle mass increase further with overload [18]. This suggests that it is the $\alpha 1$ isoform of AMPK that can block hypertrophy. Second, endurance exercise activates primarily the $\alpha 2$ isoform of AMPK [19, 20], suggesting that endurance exercise does not activate the isoform of AMPK that blocks growth. If these data are combined with those of Thomson et al. [15], who injected rats with AICAR (that activates all forms of AMPK) and blocked mTORC1 activation, the conclusion would be that activation of $\alpha 1$ AMPK may block mTORC1 activation and growth, or at least is activated concomitant with a secondary factor that does block growth.

The question then is: what is it that activates $\alpha 1$ AMPK and inhibits mTORC1 and is it a causal relationship between AMPK and mTORC1 or merely a correlation? mTORC1 is activated not only by resistance exercise but by amino acids as well [21]. The activation of mTORC1 by amino acids is particularly sensitive to the amino acid leucine (Fig. 7.1). Leucine is important since it directly activates mTORC1 through a molecular process where it binds to a protein named sestrin [22, 23]. When bound to leucine, sestrin no longer inhibits recruitment of mTOR to its activator Rheb [24]. Therefore, leucine binding to sestrin initiates the activation of mTORC1. Interestingly, overexpression of sestrin is associated with an increase in AMPK phosphorylation, while at the same time exerting its negative effect on mTORC1 [25]. Also, removal of $\alpha 1$ AMPK prevents sestrin from inhibiting mTORC1 [25]. Together, these observations suggest that regulation of sestrin activity provides a mechanism for simultaneously regulating mTORC1 and AMPK activity, with increased sestrin impairing muscle growth. So, how can sestrin activity be regulated in muscle cells? The most compelling link is that, in many cells, sestrin gene expression is regulated by the tumor suppressor protein p53 [25, 26]. p53 is known to be activated following endurance exercise [27–29], but also through cellular stressors such as fasting, immobilization, shifts in redox state, age, and excessive growth signaling [29–32]. p53 activation induces upregulation of sestrin [25, 26], and with greater amounts of sestrin in muscle, it becomes harder for leucine-rich protein to activate mTORC1, resulting in impaired growth. This represents a possible mechanism behind the interference effect. Consistent with this hypothesis, it takes more protein in each meal to increase muscle protein synthesis and maintain or grow muscle when a person is in a caloric deficit [33, 34], in bed for a period of unloading [35], or older [36, 37], all of which are associated with increased p53 activity [30–32]. Such a mechanism would argue that there is nothing unique about the concurrent training effect, rather that any

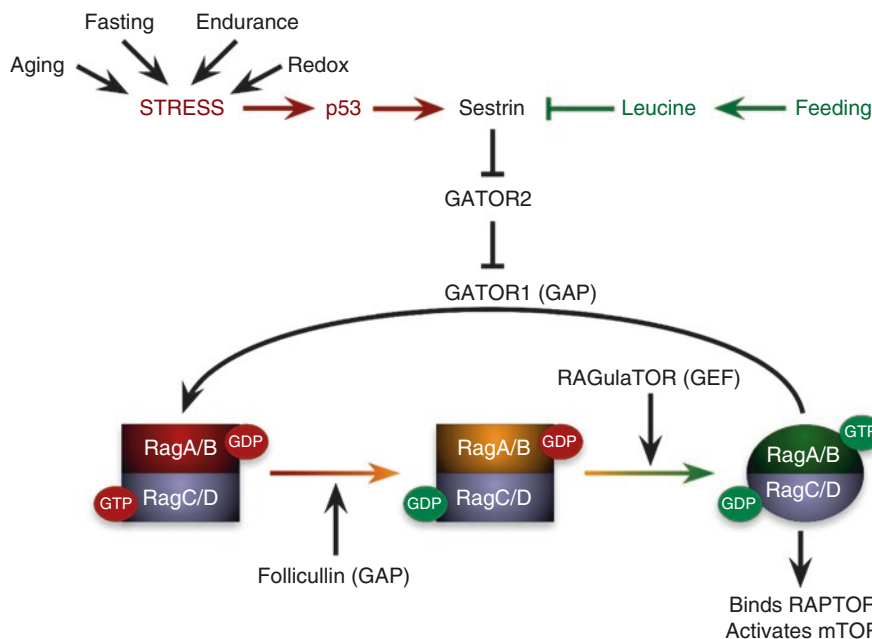


Fig. 7.1 Schematic representing the activation of mTORC1 by leucine. Here, feeding leucine-rich protein results in the inhibition of sestrin and activation of the GATOR2 complex. When activated, GATOR2 prevents GATOR1 from activating the GTPase activity of RagA/B resulting in GTP loading and recruitment of mTOR to its activator Rheb resulting in mTORC1 activation. In the presence of stress from a variety of sources, p53 is stabilized and increases sestrin levels, making it harder for leucine-rich protein to activate mTORC1

significant stress would result in a decrease in mTORC1 signaling in response to feeding, resulting in poorer muscle growth.

Beyond sestrin, p53 activation could limit muscle hypertrophy by regulating ribosome mass. Ribosome mass is positively related to the amount of muscle hypertrophy that occurs with training [38]. p53 is one of the better characterized regulators of ribosome biogenesis (Fig. 7.2). In response to stress, p53 protein is stabilized and disrupts the interaction between the ribosomal transcriptional regulators: upstream binding factor (UBF) and selectivity factor (SL1) [39]. With less interaction between UBF and SL1, the activity of polymerase I, which transcribes the ribosomal RNAs, decreases [39]. Intuitively, this would reduce muscle growth by lowering the capacity for protein translation. Consistent with a role of p53 in regulating ribosome mass, older individuals, who have more p53 activity [31], show less ribosome biogenesis and muscle hypertrophy than young individuals following resistance exercise [40].

A mechanism where the stress induced by endurance exercise underlies the interference effect would also explain why greater training volume and intensity result in greater interference. Greater training volumes and intensities would increase redox and metabolic stress within the working muscles, resulting in greater increases in p53 activity, greater sestrin activity, more difficulty activating mTORC1 in response to feeding, and lower ribosome mass resulting in lower rates of protein synthesis. Consistent with this

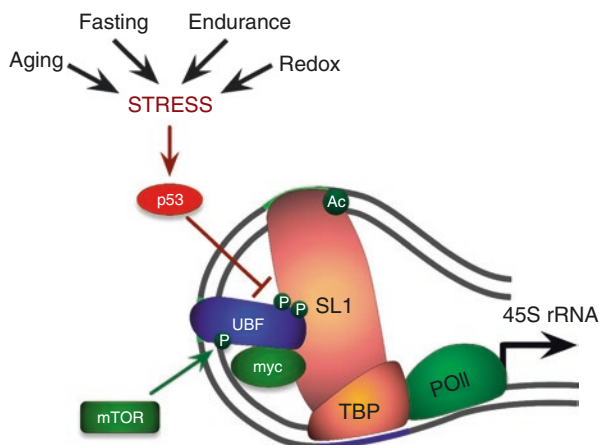


Fig. 7.2 Schematic representing the effect of stress on ribosome biogenesis. Stress from either endurance exercise, fasting, shifts in redox state, or aging can increase p53. Secondary to the increase in p53, there is destabilization of the interaction between the essential transcription factors SL1 and UBF resulting in a decrease in polymerase I activity and a decrease in production of the 45S pre-ribosomal RNA

model, equal workloads (whether low or high intensity) result in similar p53 activation [27], and increasing the metabolic stress of training by reducing carbohydrate availability results in a further increase in p53 activity [28]. This suggests that the greater the metabolic stress the greater the p53 activation and the greater the interference effect, consistent with the training data from human studies.

The Interference Effect and Muscle Power

Even though the greatest amount of attention has been paid to the limits of muscle growth, the strongest component of the interference effect is its effect on power [4]. Power is the product of force and velocity and is therefore determined by muscle cross-sectional area, the neural activation of the muscle, the isoform of myosin heavy chain expressed in the muscle, and the ability to transmit force from the myosin and actin within each sarcomere through the tendon to the bone [41]. Therefore, in theory each of these factors may show an interference effect. We discussed the potential interference effect on cross-sectional area above; however, one important aspect of the interference effect on muscle size was not discussed: the effect on fiber type. Strength training preferentially increases the amount of Type II myosin heavy chain protein in muscle [42]. Even though the number of fast fibers does not change with training, muscle growth preferential occurs through addition of Type IIa myosin heavy chain protein in existing fibers, both by increasing their cross-sectional area and by transforming IIx fibers into IIa fibers [43]. This increases the Type II myosin area and increases strength and power [5, 43, 44]. In contrast to strength training, endurance and strength performed together results in greater hypertrophy of Type I fibers [45], while simultaneously limiting growth in Type II fibers [46].

The resulting muscle would have a greater proportion of slow myosin and may therefore show a lower power than muscle trained exclusively for strength. Accordingly, muscles from athletes completing endurance training or combined endurance and strength training may demonstrate lower power than muscles from athletes trained exclusively for strength or power [47]. As for neural adaptations, strength training generally alters neural functions to facilitate activation of a greater number of muscle fibers [48]. This increases the potential for activating Type II muscle fibers during maximal efforts, providing increased muscle strength and power. These adaptations may be prone to the interference effect, as suggested by studies showing impaired alterations in voluntary activation, EMG, or rate of force development after concurrent training [49, 50]. The interference may also be ascribed residual fatigue from the endurance training, compromising the quality of strength training. However, it is beyond the scope of this chapter to elaborate on potential neurological mechanisms. Lastly, endurance exercise might exert interference by altering the microarchitecture of muscle [51]. Here, there is far less information as to how different types of training result in shifts in force transfer proteins in the muscle, within the extracellular matrix of the muscle, as well as in the tendons and therefore whether this contributes to the interference effect cannot be determined at this point.

Summary

Even though there is good empirical data for an interference effect with high volume or high intensity endurance training, the molecular mechanism has proven difficult to establish. The mechanism presented here, whereby the chronic metabolic stress of high intensity and high volume endurance exercise results in the activation of p53 and this leads to an increase in sestrin and a decrease in ribosome biogenesis, would explain much of the experimental data in the literature. Importantly, this mechanism would not only apply to the interference of endurance in the adaptation to strength training, but would also explain the difficulty in increasing muscle mass and strength in caloric restriction, unloading, and aging. Such a broadly applicable mechanism would prove easier to target using timed training and nutrition and may even prove to be a target for the development of new drugs to fight muscle wasting.

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References

1. Hickson RC. Interference of strength development by simultaneously training for strength and endurance. *Eur J Appl Physiol Occup Physiol*. 1980;45(2–3):255–63. <https://doi.org/10.1007/bf00421333>.
2. Baar K. Using molecular biology to maximize concurrent training. *Sports Med*. 2014;44(Suppl 2):S117–25. <https://doi.org/10.1007/s40279-014-0252-0>.
3. Doma K, Deakin GB. The acute effects intensity and volume of strength training on running performance. *Eur J Sport Sci*. 2014;14(2):107–15. <https://doi.org/10.1080/17461391.2012.726653>.

4. Wilson JM, Marin PJ, Rhea MR, Wilson SM, Loenneke JP, Anderson JC. Concurrent training: a meta-analysis examining interference of aerobic and resistance exercises. *J Strength Cond Res.* 2012;26(8):2293–307. <https://doi.org/10.1519/JSC.0b013e31823a3e2d>. National Strength & Conditioning Association.
5. Kraemer WJ, Patton JF, Gordon SE, Harman EA, Deschenes MR, Reynolds K, et al. Compatibility of high-intensity strength and endurance training on hormonal and skeletal muscle adaptations. *J Appl Physiol.* 1995;78(3):976–89.
6. Berryman N, Maurel DB, Bosquet L. Effect of plyometric vs. dynamic weight training on the energy cost of running. *J Strength Cond Res.* 2010;24(7):1818–25. <https://doi.org/10.1519/JSC.0b013e3181def1f5>.
7. Fyfe JJ, Bishop DJ, Stepto NK. Interference between concurrent resistance and endurance exercise: molecular bases and the role of individual training variables. *Sports Med.* 2014;44(6):743–62. <https://doi.org/10.1007/s40279-014-0162-1>.
8. Apro W, Moberg M, Hamilton DL, Ekblom B, van Hall G, Holmberg HC, Blomstrand E. Resistance exercise-induced S6K1 kinase activity is not inhibited in human skeletal muscle despite prior activation of AMPK by high-intensity interval cycling. *Am J Physiol Endocrinol Metab.* 2015;308(6):E470–81. <https://doi.org/10.1152/ajpendo.00486.2014>.
9. Apro W, Wang L, Ponten M, Blomstrand E, Sahlin K. Resistance exercise induced mTORC1 signaling is not impaired by subsequent endurance exercise in human skeletal muscle. *Am J Physiol Endocrinol Metab.* 2013;305(1):E22–32. <https://doi.org/10.1152/ajpendo.00091.2013>.
10. Coffey VG, Hawley JA. Concurrent exercise training: do opposites distract? *J Physiol.* 2017;595(9):2883–96. <https://doi.org/10.1113/JP272270>.
11. Inoki K, Zhu T, Guan KL. TSC2 mediates cellular energy response to control cell growth and survival. *Cell.* 2003;115(5):577–90.
12. Baar K, Esser K. Phosphorylation of p70(S6k) correlates with increased skeletal muscle mass following resistance exercise. *Am J Physiol.* 1999;276(1 Pt 1):C120–7.
13. Winder WW, Hardie DG. Inactivation of acetyl-CoA carboxylase and activation of AMP-activated protein kinase in muscle during exercise. *Am J Physiol.* 1996;270(2 Pt 1):E299–304.
14. Thomson DM, Gordon SE. Diminished overload-induced hypertrophy in aged fast-twitch skeletal muscle is associated with AMPK hyperphosphorylation. *J Appl Physiol.* 2005;98(2):557–64.
15. Thomson DM, Fick CA, Gordon SE. AMPK activation attenuates S6K1, 4E-BP1, and eEF2 signaling responses to high-frequency electrically stimulated skeletal muscle contractions. *J Appl Physiol.* 2008;104(3):625–32. <https://doi.org/10.1152/japplphysiol.00915.2007>.
16. Atherton PJ, Babraj J, Smith K, Singh J, Rennie MJ, Wackerhage H. Selective activation of AMPK-PGC-1 α or PKB-TSC2-mTOR signaling can explain specific adaptive responses to endurance or resistance training-like electrical muscle stimulation. *FASEB J.* 2005;19(7):786–8.
17. McGee SL, Mustard KJ, Hardie DG, Baar K. Normal hypertrophy accompanied by phosphorylation and activation of AMP-activated protein kinase α 1 following overload in LKB1 knockout mice. *J Physiol.* 2008;586(6):1731–41. <https://doi.org/10.1113/jphysiol.2007.143685>.
18. Mounier R, Lantier L, Leclerc J, Sotiropoulos A, Pende M, Daegelen D, et al. Important role for AMPK α 1 in limiting skeletal muscle cell hypertrophy. *FASEB J.* 2009;23(7):2264–73. <https://doi.org/10.1096/fj.08-119057>.
19. Lee-Young RS, Koufogiannis G, Canny BJ, McConell GK. Acute exercise does not cause sustained elevations in AMPK signaling or expression. *Med Sci Sports Exerc.* 2008;40(8):1490–4. <https://doi.org/10.1249/MSS.0b013e318173a037>.
20. Wojtaszewski JF, Nielsen P, Hansen BF, Richter EA, Kiens B. Isoform-specific and exercise intensity-dependent activation of 5'-AMP-activated protein kinase in human skeletal muscle. *J Physiol.* 2000;528(Pt 1):221–6.
21. Bar-Peled L, Sabatini DM. Regulation of mTORC1 by amino acids. *Trends Cell Biol.* 2014;24(7):400–6. <https://doi.org/10.1016/j.tcb.2014.03.003>.

22. Saxton RA, Knockenhauer KE, Wolfson RL, Chantranupong L, Pacold ME, Wang T, et al. Structural basis for leucine sensing by the Sestrin2-mTORC1 pathway. *Science*. 2016;351(6268):53–8. <https://doi.org/10.1126/science.aad2087>.
23. Wolfson RL, Chantranupong L, Saxton RA, Shen K, Scaria SM, Cantor JR, Sabatini DM. Sestrin2 is a leucine sensor for the mTORC1 pathway. *Science*. 2016;351(6268):43–8. <https://doi.org/10.1126/science.aab2674>.
24. Bar-Peled L, Chantranupong L, Cherniack AD, Chen WW, Ottina KA, Grabiner BC, et al. A Tumor suppressor complex with GAP activity for the Rag GTPases that signal amino acid sufficiency to mTORC1. *Science*. 2013;340(6136):1100–6. <https://doi.org/10.1126/science.1232044>.
25. Budanov AV, Karin M. p53 target genes sestrin1 and sestrin2 connect genotoxic stress and mTOR signaling. *Cell*. 2008;134(3):451–60. <https://doi.org/10.1016/j.cell.2008.06.028>.
26. Deng W, Cha J, Yuan J, Haraguchi H, Bartos A, Leishman E, et al. p53 coordinates decid-ual sestrin 2/AMPK/mTORC1 signaling to govern parturition timing. *J Clin Invest*. 2016;126(8):2941–54. <https://doi.org/10.1172/JCI87715>.
27. Bartlett JD, Hwa Joo C, Jeong TS, Louhelainen J, Cochran AJ, Gibala MJ, et al. Matched work high-intensity interval and continuous running induce similar increases in PGC-1alpha mRNA, AMPK, p38, and p53 phosphorylation in human skeletal muscle. *J Appl Physiol*. 2012;112(7):1135–43. <https://doi.org/10.1152/jappphysiol.01040.2011>.
28. Bartlett JD, Louhelainen J, Iqbal Z, Cochran AJ, Gibala MJ, Gregson W, et al. Reduced carbohydrate availability enhances exercise-induced p53 signaling in human skeletal muscle: implications for mitochondrial biogenesis. *Am J Physiol Regul Integr Comp Physiol*. 2013;304(6):R450–8. <https://doi.org/10.1152/ajpregu.00498.2012>.
29. Tachtsis B, Smiles WJ, Lane SC, Hawley JA, Camera DM. Acute endurance exercise induces nuclear p53 abundance in human skeletal muscle. *Front Physiol*. 2016;7:144. <https://doi.org/10.3389/fphys.2016.00144>.
30. Aquilano K, Baldelli S, Pagliei B, Cannata SM, Rotilio G, Ciriolo MR. p53 orchestrates the PGC-1alpha-mediated antioxidant response upon mild redox and metabolic imbalance. *Antioxid Redox Signal*. 2013;18(4):386–99. <https://doi.org/10.1089/ars.2012.4615>.
31. Edwards MG, Anderson RM, Yuan M, Kendzierski CM, Weindrich R, Prolla TA. Gene expression profiling of aging reveals activation of a p53-mediated transcriptional program. *BMC Genomics*. 2007;8:80. <https://doi.org/10.1186/1471-2164-8-80>.
32. Fox DK, Ebert SM, Bongers KS, Dyle MC, Bullard SA, Dierdorff JM, et al. p53 and ATF4 mediate distinct and additive pathways to skeletal muscle atrophy during limb immobilization. *Am J Physiol Endocrinol Metab*. 2014;307(3):E245–61. <https://doi.org/10.1152/ajpendo.00010.2014>.
33. Areta JL, Burke LM, Camera DM, West DW, Crawshaw S, Moore DR, et al. Reduced resting skeletal muscle protein synthesis is rescued by resistance exercise and protein ingestion following short-term energy deficit. *Am J Physiol Endocrinol Metab*. 2014;306(8):E989–97. <https://doi.org/10.1152/ajpendo.00590.2013>.
34. Garthe I, Raastad T, Refsnes PE, Koivisto A, Sundgot-Borgen J. Effect of two different weight-loss rates on body composition and strength and power-related performance in elite athletes. *Int J Sport Nutr Exerc Metab*. 2011;21(2):97–104.
35. English KL, Mettler JA, Ellison JB, Mamerow MM, Arentson-Lantz E, Patarini JM, et al. Leucine partially protects muscle mass and function during bed rest in middle-aged adults. *Am J Clin Nutr*. 2016;103(2):465–73. <https://doi.org/10.3945/ajcn.115.112359>.
36. De Bandt JP. Leucine and mammalian target of rapamycin-dependent activation of muscle protein synthesis in aging. *J Nutr*. 2016;146(12):2616S–24S. <https://doi.org/10.3945/jn.116.234518>.
37. Yang Y, Breen L, Burd NA, Hector AJ, Churchward-Venne TA, Josse AR, et al. Resistance exercise enhances myofibrillar protein synthesis with graded intakes of whey protein in older men. *Br J Nutr*. 2012;108(10):1780–8. <https://doi.org/10.1017/S0007114511007422>.
38. Stec MJ, Kelly NA, Many GM, Windham ST, Tuggle SC, Bamman MM. Ribosome biogenesis may augment resistance training-induced myofiber hypertrophy and is required for

- myotube growth in vitro. *Am J Physiol Endocrinol Metab.* 2016;310(8):E652–61. <https://doi.org/10.1152/ajpendo.00486.2015>. ajpendo 00486 02015.
39. Zhai W, Comai L. Repression of RNA polymerase I transcription by the tumor suppressor p53. *Mol Cell Biol.* 2000;20(16):5930–8.
 40. Stec MJ, Mayhew DL, Bamman MM. The effects of age and resistance loading on skeletal muscle ribosome biogenesis. *J Appl Physiol.* 2015;119(8):851–7. <https://doi.org/10.1152/jappphysiol.00489.2015>.
 41. Hughes DC, Wallace MA, Baar K. Effects of aging, exercise, and disease on force transfer in skeletal muscle. *Am J Physiol Endocrinol Metab.* 2015;309(1):E1–E10. <https://doi.org/10.1152/ajpendo.00095.2015>.
 42. Kosek DJ, Kim JS, Petrella JK, Cross JM, Bamman MM. Efficacy of 3 days/wk resistance training on myofiber hypertrophy and myogenic mechanisms in young vs. older adults. *J Appl Physiol.* 2006;101(2):531–44. <https://doi.org/10.1152/jappphysiol.01474.2005>.
 43. Staron RS, Leonardi MJ, Karapondo DL, Malicky ES, Falkel JE, Hagerman FC, Hikida RS. Strength and skeletal muscle adaptations in heavy-resistance-trained women after detraining and retraining. *J Appl Physiol.* 1991;70(2):631–40. <https://doi.org/10.1152/jappl.1991.70.2.631>.
 44. Verdijk LB, Gleeson BG, Jonkers RA, Meijer K, Savelberg HH, Dendale P, van Loon LJ. Skeletal muscle hypertrophy following resistance training is accompanied by a fiber type-specific increase in satellite cell content in elderly men. *J Gerontol A Biol Sci Med Sci.* 2009;64(3):332–9. <https://doi.org/10.1093/gerona/gln050>.
 45. Kazior Z, Willis SJ, Moberg M, Apro W, Calbet JA, Holmberg HC, Blomstrand E. Endurance exercise enhances the effect of strength training on muscle fiber size and protein expression of Akt and mTOR. *PLoS One.* 2016;11(2):e0149082. <https://doi.org/10.1371/journal.pone.0149082>.
 46. Aagaard P, Andersen JL, Bennekou M, Larsson B, Olesen JL, Crameri R, et al. Effects of resistance training on endurance capacity and muscle fiber composition in young top-level cyclists. *Scand J Med Sci Sports.* 2011;21(6):e298–307. <https://doi.org/10.1111/j.1600-0838.2010.01283.x>.
 47. Sleivert GG, Backus RD, Wenger HA. Neuromuscular differences between volleyball players, middle distance runners and untrained controls. *Int J Sports Med.* 1995;16(6):390–8. <https://doi.org/10.1055/s-2007-973026>.
 48. Gandevia SC. Spinal and supraspinal factors in human muscle fatigue. *Physiol Rev.* 2001;81(4):1725–89.
 49. Eklund D, Pulverenti T, Bankers S, Avela J, Newton R, Schumann M, Häkkinen K. Neuromuscular adaptations to different modes of combined strength and endurance training. *Int J Sports Med.* 2015;36(02):120–9. <https://doi.org/10.1055/s-0034-1385883>.
 50. Häkkinen K, Alen M, Kraemer WJ, Gorostiaga E, Izquierdo M, Rusko H, et al. Neuromuscular adaptations during concurrent strength and endurance training versus strength training. *Eur J Appl Physiol.* 2003;89(1):42–52. <https://doi.org/10.1007/s00421-002-0751-9>.
 51. Hughes DC, Ellefsen S, Baar K. Adaptations to endurance and strength training. *Cold Spring Harb Perspect Med.* 2018;8(6):a029769. <https://doi.org/10.1101/cshperspect.a029769>.